

Morphophonology

Topic 6: Phonology licensed by morphology?



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Final clusters in Mallorcan Catalan



The mystery: 1sg.PI in verbs

- At the end of a word: rising sonority consonant clusters allowed, but *only* in the 1sg.PI

Balearic Catalan	Central Catalan	
['əmpr]	['empru]	I use'
['ɔpr]	['ɔβru]	I open'
['fɪltr]	['fɪltru]	I filter'
['ʃukl]	['ʃuklu]	I sip'

A very similar distribution, found in other insular varieties of Catalan (e.g. Algherese). Here, data from Mallorcan.



Masculine nominals

- Masculine nominals surface with a final vowel, normally schwa [ə], when the root ends in the same type of clusters.

['membərə]	member'
['besprə]	evening'
['fɪltrə]	filter'
['sɪklə]	cycle'

Epenthesis and a puzzle

- The usual interpretation: final schwa in nominals is an epenthetic vowel that repairs ill-formed syllable structure.
 - Evidence for schwa epenthesis in the language: *stop*: [əs'tɒp]
- The puzzle: why doesn't epenthesis apply as needed in 1sg.PI?

If 'filter'(N) /fɪltr/ → ['fɪltrə], why not 'filter'(1sg.PI) → *['fɪltrə]



Two approaches to be considered

- Paradigmatic approaches (transderivational)
 - Based on paradigm uniformity
 - Based on paradigmatic contrast
- Alignment-based approach (prosodic)



Paradigmatic approach based on paradigm uniformity (Lloret 2004)

- Basic idea: epenthesis blocked in order to favor identity between the shared lexemes (stem) of the verbal paradigm (Optimal Paradigms model McCarthy 2005). Uniform exponence.
 - Candidates contain all the members of the verbal paradigm



Constraints

- Relevant constraints:
 - OP-Dep: penalizes shared lexemes with added segments
 - *AA: penalizes [aa], [əə] sequences
 - SSP: penalizes outputs with the “wrong” sonority profile
- Crucial assumption: epenthetic schwa is considered part of the stem.



OP applied to the Present Indicative paradigm

Verbs with OP: ['filtr] 'I filter'

/filtr/	*AA	OP-Dep-V	SSP	Dep-V
a.  <'filtr], 'filtr]əs, 'filtr]ə, fil'tr]am, fil'tr]aw, 'filtr]ən>			*	
b. <'filtrə], 'filtr]əs, 'filtr]ə, fil'tr]am, fil'tr]aw, 'filtr]ən>		5*		*
c. <'filtrə], 'filtrə]əs, 'filtrə]ə, fil'trə]am, fil'trə]aw, 'filtrə]ən>	5*!			6*

OP applied to nouns

- Nouns have a two-cell paradigm. No problem with masculine nouns (feminine forms have an inflectional vowel)

Masculine nouns with Optimal Paradigms: ['filtrə] 'filter'

/filtr/	*AA	OP-Dep-V	SSP	Dep-V
a.  <'filtrə], 'filtrə]s>				**
b. <'filtr], 'filtr]s>			**!	
c. <'filtrə], 'filtr]s>		*!		*
d. <'filtr], 'filtrə]s>		*!		*



OP in adjectives: a problem

- Adjectives have a four-cell paradigm
- Feminine forms have the regular schwa ending
 - cf. ['prim] (masc.), ['primə] (fem.)

Masculine adjectives with Optimal Paradigms: ['asprə] 'rough'

/aspr/	*AA	OP-Dep-V	SSP	Dep-V
a.  <'aspr], 'aspr]s>, 'aspr]ə, 'aspr]əs			*	
b.  <'asprə], 'asprə]s>, 'aspr]ə, 'aspr]əs		4*!		**
c. <'asprə], 'asprə]s>, 'asprə]ə, 'asprə]əs	**!			4*

Possible fixes and questions

- One possible way to try to fix the problem posed by adjectives: dividing the paradigm into subparadigms (masc. vs fem.).
- Open questions:
 - In addition to some phonology being sensitive to syntactic categories (nouns, verbs), should be sensitive to morphosyntactic features ($\pm$ feminine)?
 - How many subdivisions would be available to languages?



Paradigmatic approach based on paradigm contrast (PC) (Pons-Moll 2007)

- Basic idea: 1sg.PI surfaces without epenthesis in order to preserve the contrast with 3sg.PI.

1sg.PI: ['fɪltr] *['fɪltrə]

3sg.PI: ['fɪltrə]

- In this approach all forms of the paradigm are also compared to each other.
 - Candidate penalized by PC: {'fɪltrə} ... ['fɪltrə] ...}

Empirical issues

- In some varieties 3sg.PI also has no inflectional vowel, but still 1sg.PI surfaces without a schwa.

[ˈɔpr]: 1sg.PI, 3sg.PI

- In some varieties 3sg.PI has an inflectional ending different from schwa but epenthesis in 1sg.PI does not apply.

1sg.PI: [ˈɔpr] * [ˈɔβrə]

3sg.PI: [ˈɔβri]

- The homophony found in masculine nouns and adjectives should be accounted for differently

Nouns: [ˈfɪltrə], [ˈfɪltrəs]

Adjectives: [ˈasprə], [ˈasprəs]



An analysis with prosodic boundaries (Bonet 2026)

- Basic idea
 - an alignment constraint penalizes final epenthesis
 - rising sonority clusters surface because no other exponent is available
 - Align-R: the right edge of every morphological word must coincide with the right edge of a Prosodic Word



Align-R in action

Verbs with Align-R: ['filtr] 'I filter'

/filtr/	*Align-R	SSP	Dep-V
a.  'filtr)]		*	*
b. 'filtr)ə]	*!		



Consequences of the analysis

- The high ranking of the Align-R constraint prevents final epenthesis in nouns and adjectives.
- Reinterpretation: the schwa that appears in masculine nouns and adjectives is *not* epenthetic but a masculine allomorph.



Allomorphy in the masculine

- Most masculine nominals don't have an exponent (masc: Ø); some are [ə]

Default exponent		special exponent	
['kap]	'head'	['par-ə]	'father'
[ən'sjam]	'lettuce'	['kont-ə]	'story'
['fɜrm]	'firm'	[uni'form-ə]	'uniform'

	Align-R
<i>Filtre</i> (noun)	
'filtrə]	✗
'filtr-ə)]	✓



The proposal in more detail

- The distribution of the allomorphs is partially phonologically conditioned, with $\emptyset$ as the preferred allomorph
 - Masc $\rightarrow \{\emptyset > \textcolor{green}{\text{ə}}\}$
- The whole pack is inserted in the input to the phonology
- A constraint Priority penalizes the selection of the less preferred allomorph (Mascaró 2007).



Align-R and Priority at work

- Regular masculine selection

/fɛrm-{Ø>ə}/	*Align-R	SSP	Priority	Dep-V
a.  'fɛrm)]				
b. 'fɛrm)ə]	*!			*
c. 'fɛrmə)]			*!	

- Special masculine selection

/filtr-{Ø>ə}/	*Align-R	SSP	Priority	Dep-V
a. 'filtr)]		*!		
b. 'filtr)ə]	*!			*
c.  'filtrə)]			*	



No allomorph available in verbs

- The incorporation of Priority doesn't change the analysis of verbs.

/filtr/	*Align-R	SSP	Priority	Dep-V
a.  'filtr)]		*		
b. 'filtr) ]	*!			*



More consequences of the analysis

- Final schwa epenthesis has been commonly assumed to occur in other contexts, mostly in verbal morphology.
- The high ranking of Align-R prevents these final schwas from being interpreted as epenthetic.
- They must be reanalyzed as morphologically affiliated.
- Not here.



A final question

- Is this analysis compatible with a parallel version of OT?
 - A problem might arise with nouns and adjectives when concatenated with vowel-initial words (judgements for Balearic Catalan pending).



Verbs followed by V-initial words

‘Internal filter’:

- a. ??[_ifil.trØ)jn.'tɛrn] candidate with preferred allomorph (no VV)
- b. ?[_ifil.tr_ə)jn.'tɛrn] candidate with special allomorph (diphthongization)
- In a bare parallel approach Align-R and Priority would select the no-VV candidate, with the preferred Ø-allomorph.

Alignment plus something else?

- Within a cyclic approach, no problem.
 - ?[,fil.trə)jn.'tɜrn] faithful to inner cycle 'fil.trə
- Incorporating OO-constraints, no problem.
 - ?[,fil.trə)jn.'tɜrn] faithful to output ['fil.trə
- Accounting for variation, a different issue.



Norwegian and Danish imperatives



Norwegian codas

- In general, Norwegian allows final coda consonants (singleton and geminate)
- Also clusters of various sorts
 - "Falling sonority": NT, NS, LT, LS, RT, RS
 - "Level sonority" involving S: ST, TS
- But not "rising sonority"
 - *TR, *TN, etc.



Norwegian imperatives

- Infinitives: generally verb root + -e ([ə], [a])
- Imperatives: verb root alone

å spise	spis!	'eat'
å bruke	bruk!	'use'
å gjøre	gjør!	'do'
å greie	grei!	'manage'
å bruke	bruk!	'use'
å snakke	snakk!	'speak'
å tenke	tenk!	'think'
å følge	følg!	'follow'
å ønske	ønsk!	'wish'



Verbs with rising sonority clusters

- Verb roots ending in rising sonority clusters would create illegal codas in the imperative

<i>Infinitive</i>	<i>Expected imperative</i>	<i>Gloss</i>
åpne	åpn	'open'
kvikne	kvikn	'revive'
padle	padl	'paddle'
hindre	hindr	'impede'
handle	handl	'shop'
samle	saml	'collect'
ordne	ordn	'arrange'
sykle	sykl	'bike'
klatre	klatr	'climb'
fordre	fordr	'require'



Various solutions (Rice 2003)

- Many speakers have a gap for these forms
 - At least, in phrasal contexts without a following V
- Some speakers produce syllabic sonorants
 - ǎp[ɲ]
- Some speakers devoice sonorants after voiceless C's
 - ap[ɲ̥]



A similar situation in Danish

Basbøll (2018): "But other IMP-forms do present problems to Danish speakers, their final clusters being aberrant, i.e. structurally deviant. Whereas *skeln!* ([sɡɛlʔn])... is a perfect monosyllable, this is not the case for the reverse final cluster [nl]: *hand!* 'trade!' is normally pronounced as a disyllable (with syllabic [l]): ['hanʔl]. When an /l/ follows a voiceless consonant, as e.g. in *ras!* 'rattle!', the speaker has a choice between the disyllabic pronunciation in [sɹ̥]: ['ʁasɹ̥], or a pronunciation with voiceless /l/, viz. in [sɹ̥]: [ʁasɹ̥l]; the former choice is undoubtedly the preferred one."



Syllabic vs. devoiced in Danish

- Basbøll (1970) on imperatives *klatr* /tɐ/ 'climb', *pukl* /kl/ 'swot', and *visn* /sn/ 'wither'

"These are normally pronounced as disyllables, thus coalescing with the nouns *klatter* [kladɔ] 'blots', *pukkel* [pogɪ] 'hump', and the adjective *vissen* [vesɪ] 'withered'. If, however, they are pronounced as monosyllables, the last consonant is devoiced...: [kladɔ̥], [pogɪ̥], [vesɪ̥] (these pronunciations are quoted by Rischel...)."

- These final clusters do not otherwise occur in Danish
- Rice seems to be describing something similar in Norwegian



Underapplication of cluster phonotactics

- Like Mallorcan Catalan, a set of final clusters are tolerated only in a specific morphological form
- We need a grammar that...
 - Normally eliminates these clusters
 - Lets imperatives override or escape the restriction



Underapplication of cluster phonotactics

- Cyclic approach?
- Transderivational approach
 - Imperative is faithful to infinitive
 - BD-Max(C), BD-Dep(syl)



Transderivational approach

- Hypothetical monomorphemic /tapr/

/tapr/	BD-Max(C)	BD-Dep(syl)	*TR	IO-Max(C)	IO-Dep(V)
a. tapr					
b. tap				*!	
c.  ta.pər					*
d. tapṛ			*!		

- Imperative /klatr/ (base: [klatrə])

/klatr/	BD-Max(C)	BD-Dep(syl)	*TR	IO-Max(C)	IO-Dep(syl)
a. klatr					
b. klat	*!			*	
c. kla.tṛ		*!			*
d.  klatṛ			*		



Epenthesis in Yiddish rN clusters



rN clusters in Yiddish

- Yiddish allow rN clusters before a vowel, but not before a consonant or finally ("in a coda")
- Alternations between derivationally related forms

ʃturəm `storm'

ʃturmɪʃ `stormy'

ʃirəm `umbrella'

ʃirmə `screen'

turəm `tower'

turmə `prison'

liarəm `loud noise'

liarmɪk `noisy'

forəm `form'

formə `form'

vərəm `warm'

varməs `dinner'



Overapplication in verbs

- If rN] would be in a coda in the 1sg (etc.), it is required in plural and infinitive, too (*overapplication*)
- If the rN cluster is before a vowel in 1sg (etc.), no epenthesis required (normal non-application)

1sg	ʃtu rəm	al r mir
2sg	ʃtu rəm st	al r mirst
3sg	ʃtu rəm t	al r mirt
1pl	ʃtu rəmən	al r mirŋ
2pl	ʃtu rəm t	al r mirt
3pl	ʃtu rəmən	al r mirŋ
Infin.	ʃtu rəmən	al r mirŋ



Overapplication in verbs

- Many verbs with ...rən, none with ...rN
 - varəmən `to warm up', baʃirəmən `to shield',
turəmən `to tower', furəmən `to form'
- Many are derivationally related to nouns or adjectives with rN, confirming that these verbs have undergone epenthesis
- Lack of rN is not an accident
 - We need our grammar to predict epenthesis at the end of verb stems



Overapplication in verbs

- Unexpected epenthesis in 1pl **fturəmən**
- Cyclic evaluation
 - /fturm/ → [fturəm] → /fturəm+ɨ/ → [fturəmən]
- Transderivational faithfulness
 - 1pl [fturəmən] faithful to 1sg [fturəm] (Max(V))
- Prosody?
 - [fturəm]_{PrWd}ən



Underapplication in nouns

- Nouns sometimes fail to undergo epenthesis
- This option is only available for nouns with plural -ən
 - Epenthesis not required in plural

fɪrəm	fɪrəms	`umbrella'	liar(ə)m	liarəms ~ liarmən	`loud noise'
fʊvərəm	fʊvərəms	`swarm'	for(ə)m	formən	`form'
vɔrəm	vɔrəms	`worm'	unifor(ə)m	uniformən	`uniform'
ɔrəm	ɔrəms	`arm'	farm	farmən	`farm'
tʊrəm	tʊrəms	`tower'	reform	reformən	`reform'
alərəm	alərəms	`alarm'			



Underapplication in nouns

- Unexpected lack of epenthesis in sg. [lɪarm]
- Cyclic evaluation
 - No stage in derivation of singular noun when phonological form of plural suffix is available
 - Vowel in plural [lɪarmə̃n] is epenthetic, requires access to surface form
- Transderivational faithfulness
 - Singular faithful to plural [lɪarmən] (OO-Dep(V))

